We claim:

1. A pipeline traversing robot (100) comprising:

a base (15) having three arms extending radially outward from a central hub, forming a three-armed symmetric structure configured at 120 degrees from one another to wrap around the exterior surface of a pipeline such that wheels (10) mounted at the terminal ends of each arm maintain simultaneous contact with the pipe surface from multiple radial directions;

~~plurality of screws(11) attached to the joints;~~a plurality of screws (11) positioned at the terminal ends of each arm of the base (15) ;

~~plurality of magnets(3);~~a plurality of neodymium magnets (3) fixed on the underside of the base (15) at a predetermined clearance from the pipe surface, configured to generate magnetic adhesion force to maintain stable contact with the exterior surface of metallic gas pipes during both horizontal and vertical traversal;

a motor driver (8) ~~plurality of motors (7) attached to their respective wheel (10);~~

comprising of a plurality of motors (7), each electrically connected to the motor driver (8) and mechanically coupled to a respective wheel (10) being configured to provide directional movement of the robot along the pipeline;

a castor wheel (9)~~;~~ configured for 360-degree rotational maneuverability of the robot on the pipeline surface;

~~a torch (14);~~

~~a gas sensor (2);~~

~~a temperature and humidity sensor (6)~~

~~an ultrasonic sensor (5);~~

a gas sensor (2), a temperature and humidity sensor (6), an ultrasonic sensor (5) and torch (14) ~~;~~ mounted on the base (15)

an imaging device (4)~~;~~ mounted on the base (15) and configured to capture image data of the surrounding pipeline environment for remote visual inspection;

a monitoring and control center (12);

a data store (13); and

a processing unit (1)~~;~~ electrically connected to the gas sensor (2), temperature and humidity sensor (6), ultrasonic sensor (5), imaging device (4), motor driver (8), and torch (14), and configured to receive sensor data, process the data stored in the data store (13), and transmit leakage status and alerts to the monitoring and control center (12).

~~wherein the ultrasonic sensor (5), motor driver (8), imaging device (4), torch (14), gas sensor (2), and temperature and humidity sensor (6) are mounted on a base (15) having a fidget spinner design, and are electrically connected to the processing unit (1);~~

~~wherein screws (11) are positioned at the terminal ends of the base (15);~~

~~wherein the angular adjustments facilitated by screws (11) allow the accommodation of pipes with varying diameters;~~

~~wherein the magnetic assembly (3) is fixed on the rear side of the base, maintaining sufficient distance from touching the pipes;~~

~~wherein the motor driver (8) is electrically connected to the motor (7), and the motor (7) is mechanically coupled to the wheel (10);~~

~~wherein the imaging device (4) is mounted on the robot in a position configured for environmental visual data acquisition.~~

2. A pipeline traversing robot (100) as claimed in claim 1, ~~wherein the wherein the magnetic assembly (3) is configured to generate sufficient magnetic force for stable adherence to metallic surfaces.~~

wherein the plurality of neodymium magnets (3) are fixed on the underside of each arm of the base (15) proximal to the terminal end thereof, such that the magnetic attraction force generated between the magnets (3) and the exterior metallic surface of the gas pipe maintains inward radial contact pressure of the wheels (10) against the pipe surface , configured for adhesion of robot to the pipe along with continuous traction of the wheels (10) during traversal.

3. ~~A pipeline traversing robot (100) as claimed in claim 1, wherein~~ the surface obstacles like joints and bends are traversed.

4. A pipeline traversing robot (100) as claimed in claim 1, wherein ~~the claws adjust based on the pipe diameter.~~

the plurality of screws (11) positioned at the terminal ends of the three arms forms a wrapping grip configuration around the exterior surface of pipes having diameters in the range of 150 mm to 200mm.

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